

Progress Update: Personal Knowledge Base (Second Brain)

Work Completed

Today, the initial backend setup of the Personal Knowledge Base (Second Brain) project was successfully completed using Node.js, Express.js, MongoDB, and Mongoose.

The project structure was created following a modular architecture consisting of separate folders for configuration, models, controllers, routes, middleware, services, and documentation. MongoDB was connected locally through MongoDB Compass, and a dedicated database named **secondbrain** was created.

A Note data model was designed using Mongoose to store information such as title, content, category, tags, favorite status, and timestamps. The backend server was configured using Express.js and environment variables were managed through the dotenv package.

The first REST API endpoint was implemented:

POST /api/notes

This endpoint allows users to create and store notes in the MongoDB database. The request is processed through Express routes and controllers, validated by the Mongoose model, and then persisted in MongoDB.

Proof of Functionality

The attached Postman screenshot demonstrates successful execution of the Create Note API. The API received a JSON request containing note details and returned a **201 Created** response along with the stored document.

The response confirms:

- Express server is running successfully.
- MongoDB database connection is working.
- Mongoose schema is functioning correctly.
- REST API routing is operational.
- Data is being inserted into MongoDB collections.
- JSON responses are being returned to the client.

Current Status

Completed modules:

- Project Initialization

- MongoDB Connection
- Express Server Setup
- Environment Configuration
- Note Model Creation
- Controller Layer Setup
- Route Layer Setup
- Create Note REST API
- Postman API Testing

The project currently supports note creation and storage through REST APIs. Additional CRUD operations, search functionality, category management, tag management, analytics, and Swagger documentation will be implemented in subsequent development phases.

